

Lesson 3

Understanding Physical and Network Security Concepts



Topic 3A

Understand Physical Security Concepts

Exam Objectives Covered

- 3.2 Summarize physical security concepts

Physical Security—An Outside in Approach

- Building and Property Security
- Perimeter Security
- Interior Data Center
- Network Operations Center/Server Room
- Physical Server Security

Building and Property Security: Bollards



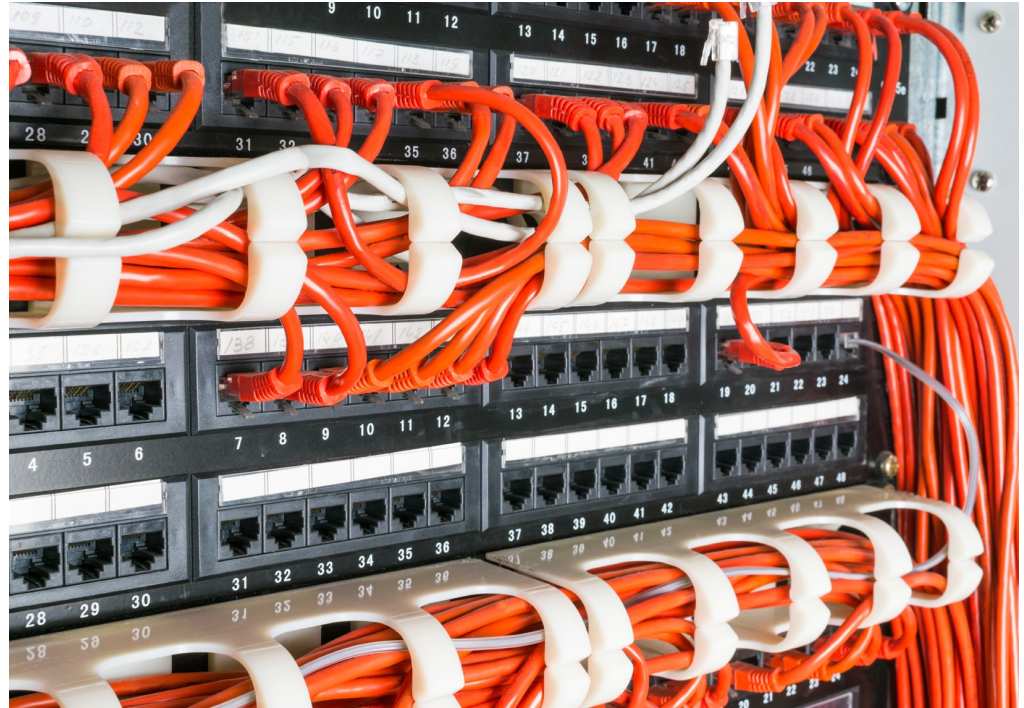
Bollards separate areas of motor traffic from buildings and pedestrian walkways. Some are powered, featuring lights and retractability, while other designs are static concrete or steel posts. (Image by Nikita Bylugin © 123RF.com)

Perimeter Security

- Exterior entry
- Interior entry
- Turnstiles
- Mantraps

Interior Data Center

- Restricted access
- Network closets
- HVAC
 - Temperature
 - Humidity
- Redundancy
- Generators

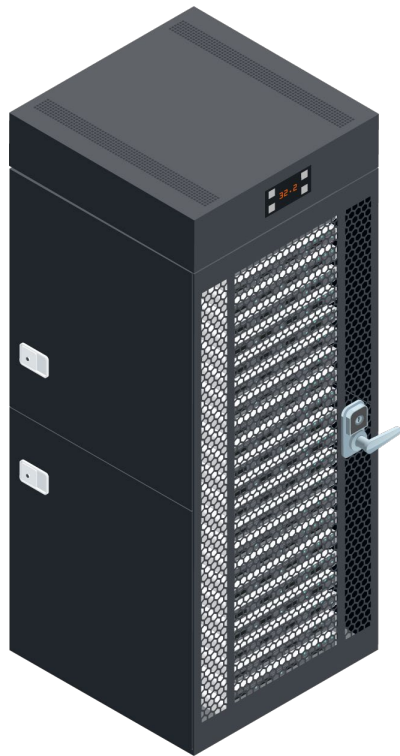


Patch panel in a network closet. (Image by Svetlana Kurochlina © 123RF.com)

Network Operations Center/Server Room

- Locks
 - Biometric scanners
 - RFID
 - Card readers
 - Electronic keyless locks
 - MFA
- Fire suppression
- Secure storage for backup media

Physical Server Security



Rack-mounted servers behind a locked door. (Image © 123RF.com)

Review Activity: Physical Security Concepts



Review Activity

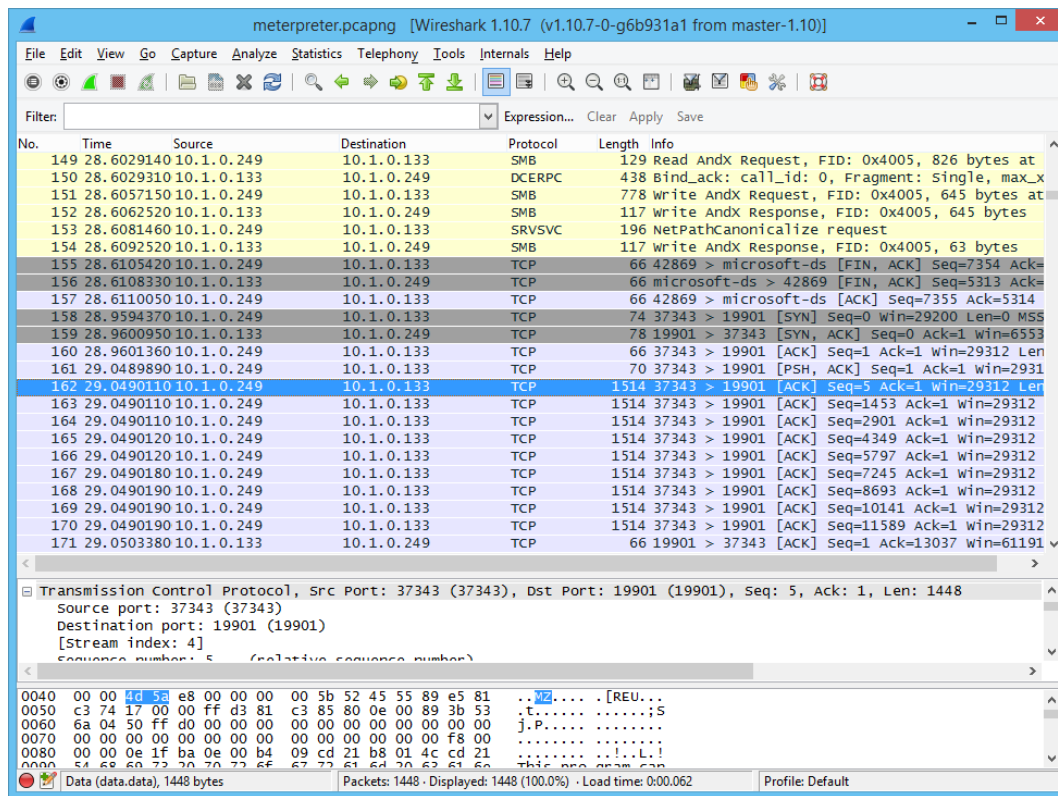
Topic 3B

Understand Network Security Groups

Exam Objectives Covered

- 2.2 Given a scenario, configure servers to use network infrastructure services

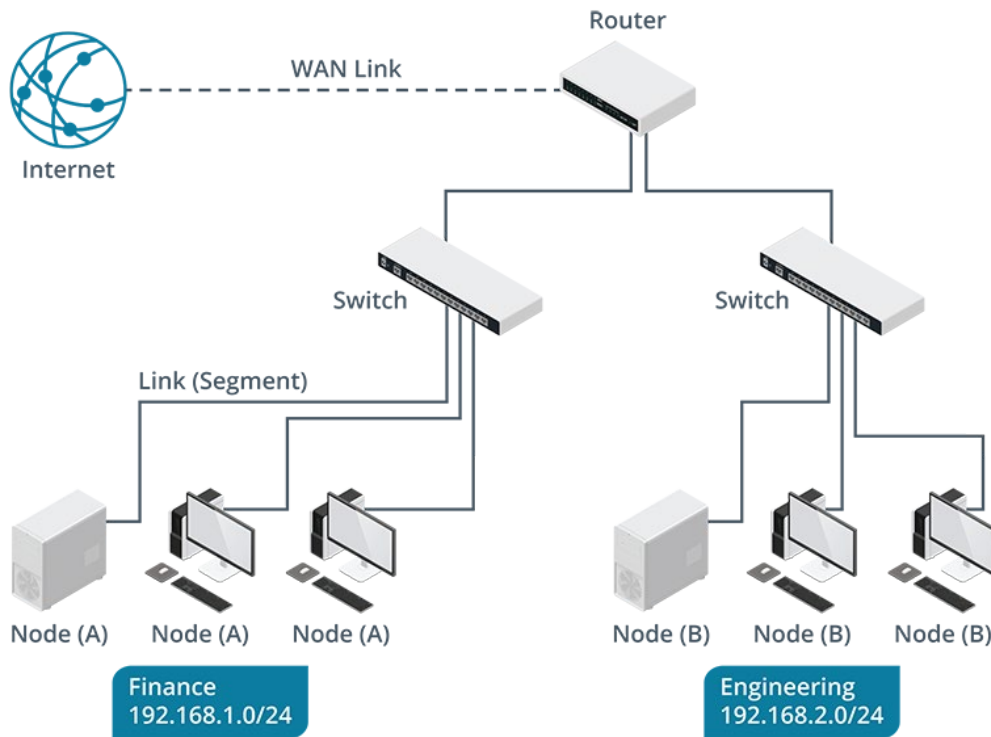
Intercept Network Traffic



Wireshark packet capturing utility. (Screenshot courtesy of Wireshark.)

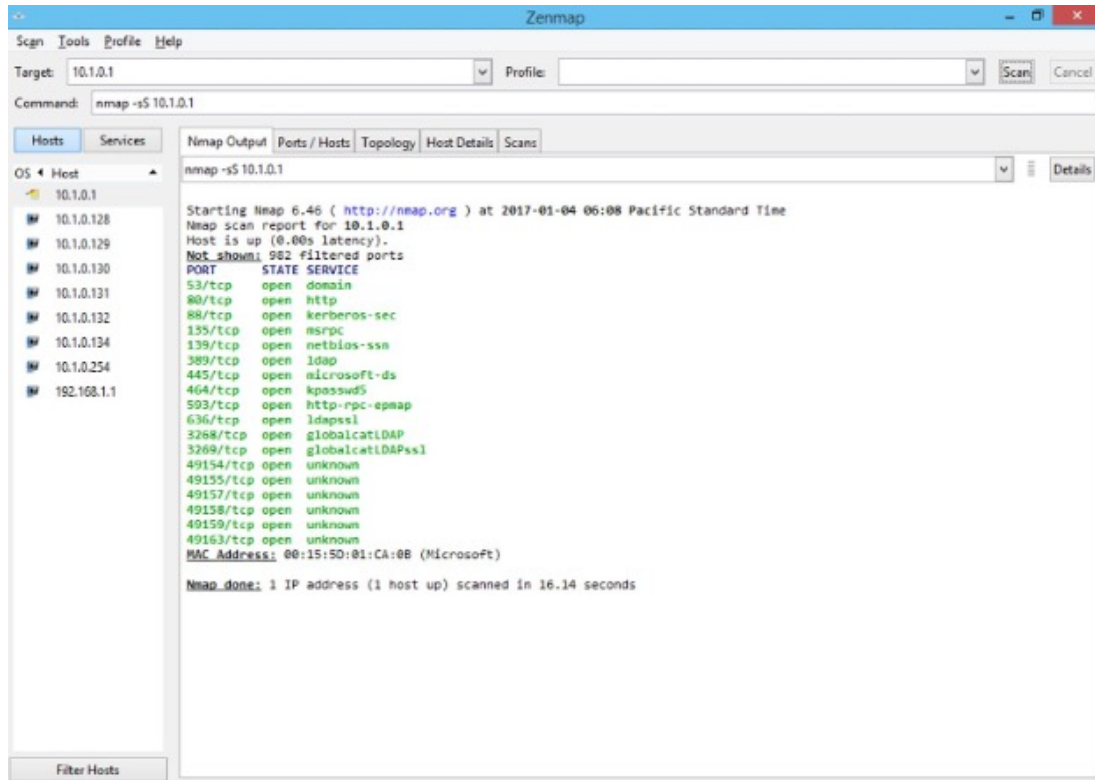
Segment Networks for Security

- DMZ
- Guest network
- Air gap



Two segments linked by a router. (Images © 123RF.com)

Network Maps



Results of an nmap scan. (Screenshot courtesy of Zenmap.)

IP Address Review

- Network ID
- Host ID

Subnet Masks

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator> ipconfig

Windows IP Configuration

Ethernet adapter vEthernet (Microsoft Hyper-V Network Adapter - Virtual Switch):

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::f15a:4fb6:327c:e5ab%20
    IPv4 Address. . . . . : 192.168.1.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1:1%20
                                192.168.1.1

Ethernet adapter vEthernet (Microsoft Hyper-V Network Adapter #2 - Virtual Switch):

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::1e1:c5a1:e744:1f44%19
    IPv4 Address. . . . . : 192.168.1.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1:1%19
                                192.168.1.1

PS C:\Users\Administrator>
```

Output from ipconfig command. Note the Subnet Mask value. (Screenshot courtesy of Microsoft.)

Reserved IP Address Ranges

- Class A, B, and C most common
- Class D and E less common

Subnetting

- Routers
- Switches

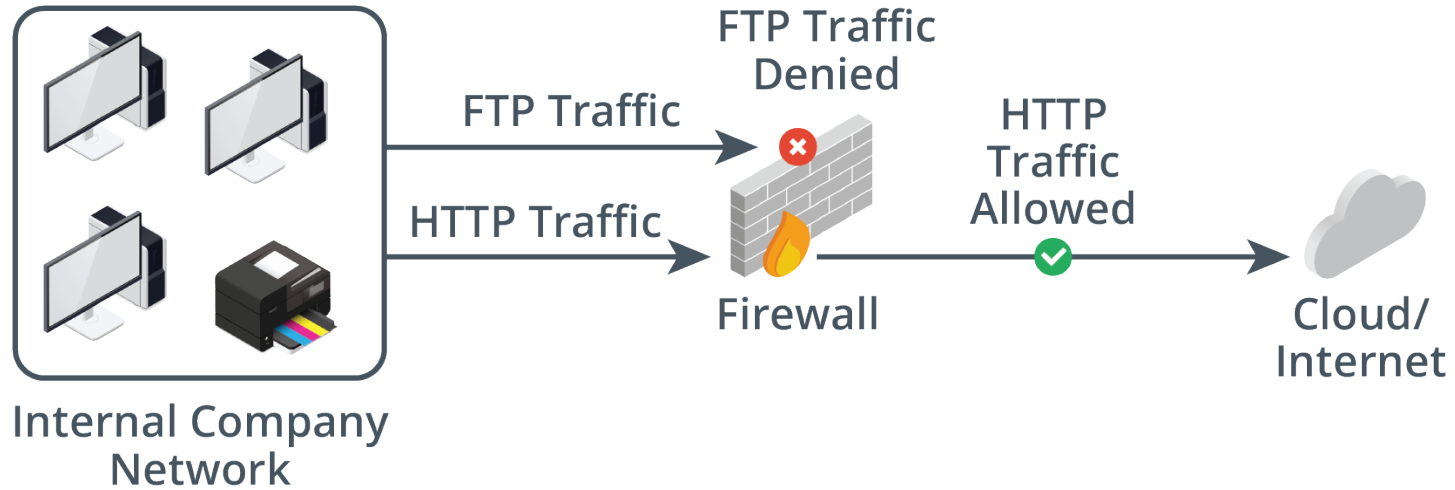
Network Hardware, Protocols, and Tools

- Servers live on a network
- Segmentation

Network Devices

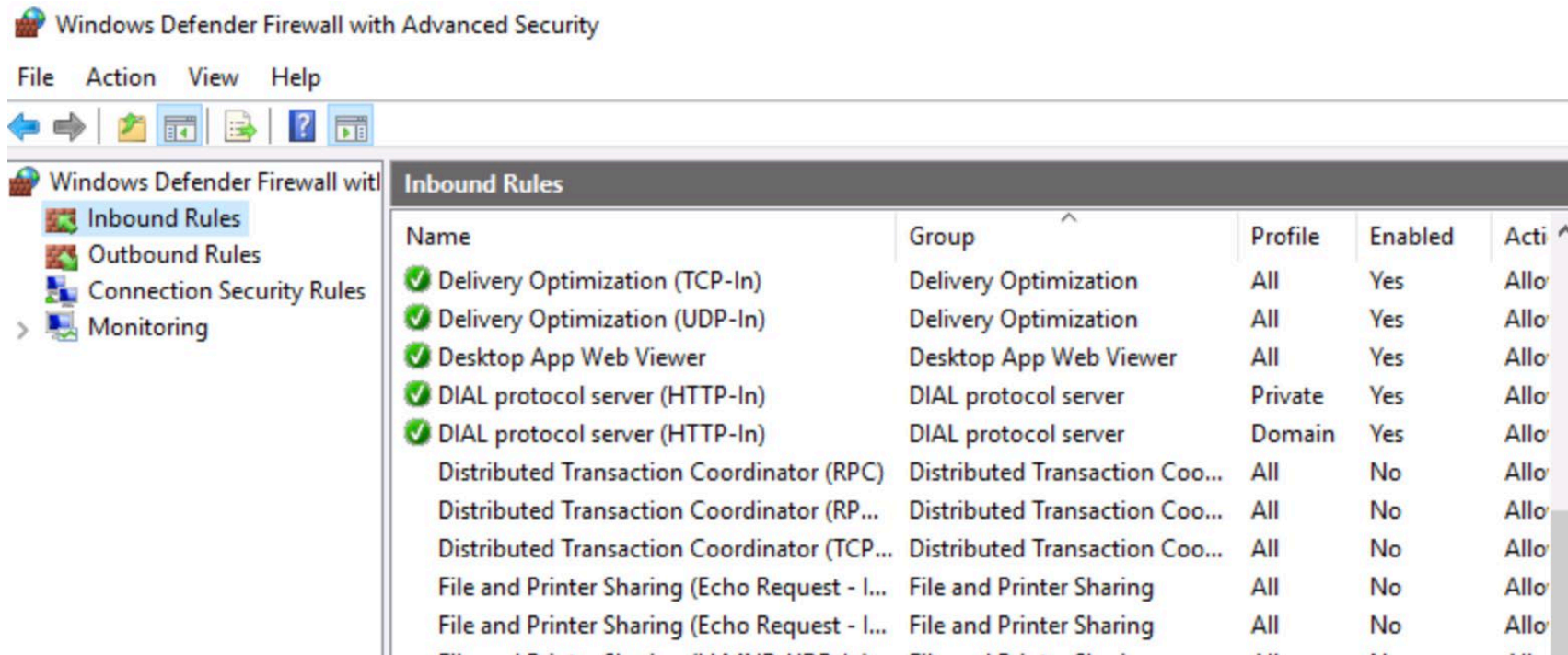
- Routers
- Switches
- Gateways
- WAPs
- Media

Network Firewalls



Network firewall separating an internal LAN and the Internet. (Images © 123RF.com)

Host-Based Firewalls



Windows Defender Firewall with Advanced Security. (Screenshot courtesy of Microsoft.)

Network Protocols

- TCP/IP
- Protocols
- Uses
- Port numbers

```
[user01@CentOS7-A ~]$ ssh root@192.168.1.100
root@192.168.1.100's password:
Last login: Mon Aug  3 15:15:09 2020 from 172.20.0.2
[root@CentOS7-A ~]# █
```

SSH Connection to 192.168.1.100

Network Access Control (NAC)

- Defines rules
- Enforces rules
- Network isolation

Review Activity: Network Security Concepts



Review Activity

Lesson 3

Summary

